Algebra and Functions

PATHWAYS
COURSE OUTLINE
WINTER 2026

General Information.

Discipline: Mathematics *Course code*: 201-016-RE
Ponderation: 2-3-3 *Credits*: $2\,^2/_3$
Number of class hours: 75
Prerequisite:
CST4, CST5 or equivalent, or a failure in TS4 or equivalent
Objective:
01PP: Solve problems by using concepts in algebra and geometry

Your teacher will provide you with contact information, a listing of office hours, and the breakdown of the class mark in your section (see the written supplement to this course outline).

Introduction. This course is designed for students who need to review or relearn basic algebraic concepts and skills. Successful completion of this course will allow you to take 201-015-AB (Algebra and Trigonometry).

Textbook. There is no required textbook for this course. Exercise sheets will be available (see links below).

Course Costs. Your instructor may recommend purchase of a scientific calculator (about $25).

Calculators are not allowed on the final exam for this course, but students planning to take other math courses should keep in mind that according to math department policy, only calculators with model number beginning with SHARP EL-531 are permitted on tests and final exams.

Teaching Methods. Classes are primarily lectures with some discussion and problem-solving. Homework normally amounts to about three hours a week. Many of the failures in this course are due to missing classes. You are expected to attend all lectures, to read the textbook, and to do the homework.

Evaluation Plan. The Final Evaluation in this course consists of the Final Exam, which covers all elements of the competency. The Final Grade is a combination of the Class Mark and the mark on the Final Exam. The Class Mark will include the student's results from three or more in-class written tests (worth at least 75% of the Class Mark), and possibly homework, quizzes and other assignments. The specifics of the Class Mark will be given by your instructor during the first week of classes in an appendix to this outline.

The Final Exam is set by the course committee, which consists of all instructors currently teaching this course, and is marked by each individual instructor.

No calculators will be allowed on the final exam.

The Final Grade will be the better of:

50% Class Mark and 50% Final Exam Mark
or
25% Class Mark and 75% Final Exam Mark

A student *choosing not to write* the Final Exam will receive a failing grade of 50% or their Class Mark, whichever is less.

Students must be available until the end of the final examination period to write exams.

Note that in the event of unexpected changes to the academic calendar, the evaluation plan may be modified.

Other Resources.

Math Website.
http://departments.johnabbott.qc.ca/departments/mathematics

Math Study Area. Located in H-200A and H-200B; the common area is usually open from 8:30 to 17:30 on weekdays as a quiet study space. Computers and printers are available for math-related assignments. It is also possible to borrow course materials when the attendant is present.

Math Help Centre. Located in H-216; teachers are on duty from 8:30 until 15:30 to give math help on a drop-in basis.

Peer Tutoring. Starting on the fifth week of each semester, first year students can be paired with a fellow finishing student for a weekly appointment of tutoring. Ask your teacher for details.

Academic Success Centre. The Academic Success Centre, located in H-139, offers study skills workshops and individual tutoring.

Selected Exercises. The exercises packages below should help you practice and learn the material taught in this course; they form a good basis for homework but they don't set a limit on the type of question that may be asked. Your teacher may supplement this list during the semester. Regular work done as the course progresses should make it easier for you to master the course. The complete list can be found here.

Order of Operations
Simplifying Polynomial Expressions
Exponent Rules
Lines
Linear Equations
Solving Linear Systems by Substitution and Elimination
Markups and Discounts
Simple Interest
Factoring
Solving by Factoring First
Radicals
Distance and Midpoint
Rationalizing the Denominator
Quadratic Equations
Equations Involving Radicals
Logarithms Exponential Equations
Trigonometry
Functions
Advanced Problems

College Policies.

Policy No. 7 - IPESA, Institutional Policy on the Evaluation of Student Achievement: https://www.johnabbott.qc.ca/wp-content/uploads/2021/05/Policy-No.-7-IPESA-FINAL.pdf.

Changes to Evaluation Plan in Course Outline (Article 5.3). Changes require documented unanimous consent from regularly attending students and approval by the department and the program dean.

Religious Holidays (Article 3.2.13 and 4.1.6). Students who wish to miss classes in order to observe religious holidays must inform their teacher of their intent in writing within the first two weeks of the semester.

Student Rights and Responsibilities: (Article 3.2.18). It is the responsibility of students to keep all assessed material returned to them and/or all digital work submitted to the teacher in the event of a grade review. (The deadline for a Grade Review is 4 weeks after the start of the next regular semester.)

Student Rights and Responsibilities: (Article 3.3.6). Students have the right to receive graded evaluations, for regular day division courses, within two weeks after the due date or exam/test date, except in extenuating circumstances. A maximum of three (3) weeks may apply in certain circumstances (ex. major essays) if approved by the department and stated on the course outline. For evaluations at the end of the semester/course, the results must be given to the student by the grade submission deadline (see current Academic Calendar). For intensive courses (i.e.: intersession, abridged courses) and AEC courses, timely feedback must be adjusted accordingly.

Academic Procedure: Academic Integrity, Cheating and Plagiarism (Article 9.1 and 9.2). Cheating and plagiarism are unacceptable at John Abbott College. They represent infractions against academic integrity. Students are expected to conduct themselves accordingly and must be responsible for all of their actions. For more information, see the Academic Procedure on Academic Integrity.

College definition of Cheating: Cheating means any dishonest or deceptive practice relative to examinations, tests, quizzes, lab assignments, research papers or other forms of evaluation tasks. Cheating includes, but is not restricted to, making use of or being in possession of unauthorized material or devices and/or obtaining or providing unauthorized assistance in writing examinations, papers or any other evaluation task and submitting the same work in more than one course without the teacher's permission. It is incumbent upon the department through the teacher to ensure students are forewarned about unauthorized material, devices or practices that are not permitted.

College definition of Plagiarism: Plagiarism is a form of cheating. It includes copying or paraphrasing (expressing the ideas of someone else in one's own words), of another person's work or the use of another person's work or ideas without acknowledgement of its source. Plagiarism can be from any source including books, magazines, electronic or photographic media or another student's paper or work.

Policy No. 13 - Policy on Student Conduct and Discipline Procedures:
https://www.johnabbott.qc.ca/wp-content/uploads/2021/09/POLICY-13_-_policy_on_student_conduct-Final-BOG-2021-09-21.pdf.

OBJECTIVES	STANDARDS
Statement of the Competency 01PP: Solve problems by using concepts in algebra and geometry.	**Performance Criterion (for the competency as a whole)** Use of appropriate terminology
Elements of the Competency	**Performance Criteria**
1. *Manipulate numerical and algebraic expressions*	1.1 Appropriate factoring of algebraic expressions (finding the common factor and factoring by grouping, perfect square trinomial, difference of two squares) 1.2 Correct application of the properties of exponents and radicals
2. *Analyze situations by using real functions.*	2.1 Appropriate modelling of the situation 2.2 Appropriate use of the multiplicative parameters 2.3 Correct determination of the properties (domain, range, sign, variation, extrema, x- and y-intercepts) of an exponential, second-degree polynomial or piecewise function 2.4 Interpretation and graphical representation of a periodic function and of the inverse of an exponential function or second degree polynomial function 2.5 Accurate interpretation of results
3. *Solve problems by using equations.*	3.1 Appropriate modelling of the problem 3.2 Correct application of the methods for solving second-degree equations (factoring and zero product rule) 3.3 Correct application of the methods for solving exponential equations, with or without logarithms (definition and changing bases) 3.4 Correct application of the methods for solving systems of linear equations in two variables 3.5 Accurate interpretation of the results
4. *Solve problems by using analytic geometry.*	4.1 Appropriate modelling of the problem 4.2 Proper use of trigonometric relations in triangles 4.3 Correct determination of the relative position of two straight lines, the equation of a line, the distance between two points and the coordinates of a point of division 4.4 Accurate interpretation of results